GPU-Accelerated Lunar Path Detection Using Modified Theta*

After detecting probable ice-rich regions and selecting safe landing zones, the rover requires a safe traverse path over the lunar DEM. The path cannot be selected only by shortest distance, because the shortest route may cross steep slopes, crater rims, boulder fields, shadowed zones, or thermally risky terrain.

Core Idea

We generate a multi-layer lunar cost field and extract the lowest-risk smooth path using modified Theta*.

Method

1. Lunar DEM is converted into a terrain grid.
2. Each terrain node is initially assigned infinite traversal cost.
3. Cost values are relaxed using terrain and mission factors.
4. Safer terrain receives lower cost; hazardous regions receive high cost or are blocked.
5. FMM/FIM-based cost propagation refines the terrain cost field.
6. Cost-field computation is parallelized across GPU cores.
7. Modified Theta* extracts a smooth line-of-sight path over the lowest-cost terrain.

Cost Factors Used

Factor	Effect on Path
Slope	Avoids steep and unstable regions
Elevation change	Reduces energy-heavy climbs and descents
Roughness	Avoids uneven, low-traction terrain
Obstacles / boulders	Blocks or penalizes unsafe regions
Crater rims	Avoids sharp terrain discontinuities
Shadowed zones	Reduces visibility and thermal risk
Illumination	Favours energy-supportive routes where relevant
Ice probability	Pulls the route toward scientifically valuable targets

Prototype Result

Tested on lunar DEM-based terrain input, the optimized approach reduced path generation time from approximately:

15 minutes → 20 seconds

Output

A smooth, low-risk, terrain-aware rover traverse path from the selected landing zone to the detected ice-rich target.